

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. :43	Name:nikita tiwari
Class :syit	Batch :
Date of Assignment :	Date/Time of Submission :

Questions:

2a. Write a python code to create nested tuples.

CODE:

```
practical no pds 1.py - C:/Users/info/AppData/Local/Programs/Python/Python314/practical no pds 1.py (3.14.6)
File Edit Format Run Options Window Help
student = ("NIKITA", 43, ("BSCIT", "SEMESTER III"), 90)
print("Name = ", student[0])
print("Roll no = ", student[1])
print("Department = ", student[2][0])
print("Semester = ", student[2][1])
print("Percentage = ", student[3])
```

OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v
.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/info/AppData/Local/Programs/Python/Python314/p
ractical no pds 1.py
Name = NIKITA
Roll no = 43
Department = BSCIT
Semester = SEMESTER III
Percentage = 90
```

2b. Write a python code to sort the nested tuple using sorted () function.

CODE:

```
practical no pds 1.py - C:/Users/info/AppData/Local/Programs/Python/Python314/practical no pds 1.py (3
File Edit Format Run Options Window Help
data = [(1,4),(3,2),(5,1)]
sorted_data = sorted(data, key = lambda x: x[1])
print(sorted_data)
```

OUTPUT:

```
= RESTART: C:/Users/info/AppData/Local/Progr
ractical no pds 1.py
[(5, 1), (3, 2), (1, 4)]
```

2c. Write a python code to copy or clone list.

CODE:

```
practical no pds 1.py - C:/Users/info/AppData/Local/Programs/Python/Python314/practical no p
File Edit Format Run Options Window Help
stud = ("NIKITA",43,("BscIT","Semester III"),85)
stud2 = ()
stud2 = stud

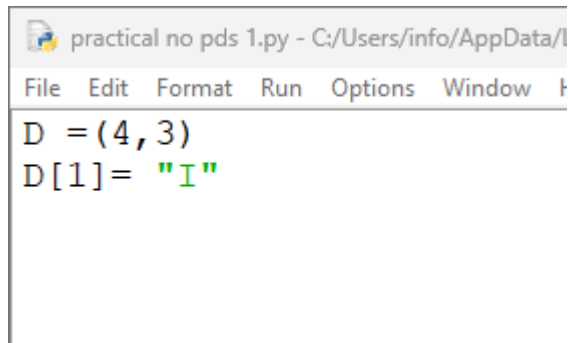
print("Name:",stud2[0])
print("Roll no.:",stud2[1])
print("Department:",stud2[2][0])
print("Semester:",stud2[2][1])
print("Percentage:",stud2[-1])
```

OUTPUT:

```
= RESTART: C:/Users/info/AppData/Lo
ractical no pds 1.py
Name: NIKITA
Roll no.: 43
Department: BscIT
Semester: Semester III
Percentage: 85
```

2d. Write a python code to check immutability property of python tuples.

CODE:



```
practical no pds 1.py - C:/Users/info/AppData/L
File Edit Format Run Options Window
D =(4, 3)
D[1]= "I"
```

```
= RESTART: C:/Users/info/AppData/Local/Programs/Python/Python314
practical no pds 1.py
Traceback (most recent call last):
  File "C:/Users/info/AppData/Local/Programs/Python/Python314/pr
tical no pds 1.py", line 2, in <module>
    D[1]= "I"
TypeError: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80